Practical 3: Visualisation using `qplot()`

Data Science (CDA) — Academic Lab Report

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Abstract

This report documents the implementation, graphical validation, and clinical data interpretation of Practical 3 (*Visualisation using `qplot()`*) for the Data Science curriculum at ETSINF (UPV). Using the `ggplot2` framework in R, we analyze real-world surgical records from $N = 394$ patients diagnosed with keratoconus who underwent intrastromal corneal ring segment (ICRS) implantation. We evaluate corneal topography parameters (primary keratometry K_1 and orthogonal curvature K_2), post-operative outcomes ($K_{1.\text{output}}$), and surgical design choices (number of rings, incision diameter, and ring thickness). The experimental pipeline covers non-parametric local polynomial smoothing (LOESS), ordinary least squares (OLS) linear regressions, aesthetic mappings, conditional discrete distributions, multi-panel faceting, and interquartile dispersion via boxplots.

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1 Introduction and Clinical Context

Keratoconus is a progressive non-inflammatory ectatic corneal dystrophy characterized by localized thinning and biomechanical weakening of corneal collagen fibrils. This structural degeneration induces a conical protrusion of the optical dome, yielding severe irregular astigmatism, high-order visual aberrations, and significant visual acuity loss.

One prominent conservative surgical intervention consists of the placement of intrastromal corneal ring segments (ICRS). These polymethyl methacrylate (PMMA) micro-segments induce an arc-shortening mechanical effect across the corneal lamellae, producing an intentional biomechanical flattening and reshaping of the anterior curvature. The dataset `Queratocono.csv` compiles clinical pre-operative measurements, surgical procedure specifications, and post-operative evaluations for 394 treated eyes across 16 numerical and categorical features.

2 Exercise 1: Curvature Relation (K_1 vs. K_2) with Smoother

2.1 Objective and Implementation

The objective is to explore the baseline anatomical dependency between the principal meridian curvature (K_1) and its orthogonal counterpart (K_2), contrasting non-parametric local polynomial regression (LOESS) against parametric linear modeling (ordinary least squares).

```
library(ggplot2)
queratocono <- read.csv("dataset/Queratocono.csv", header = TRUE, sep = ",",
    check.names = TRUE)
```

```
# 1.a) Default non-parametric smoother (LOESS)
p1_loess <- qplot(K1, K2, data = queratocono, geom = c("point", "smooth"))

# 1.b) Parametric linear regression smoother (method = "lm")
p1_lm <- qplot(K1, K2, data = queratocono, geom = c("point", "smooth"), method
    = "lm")
```

2.2 Graphical Evaluation and Discussion

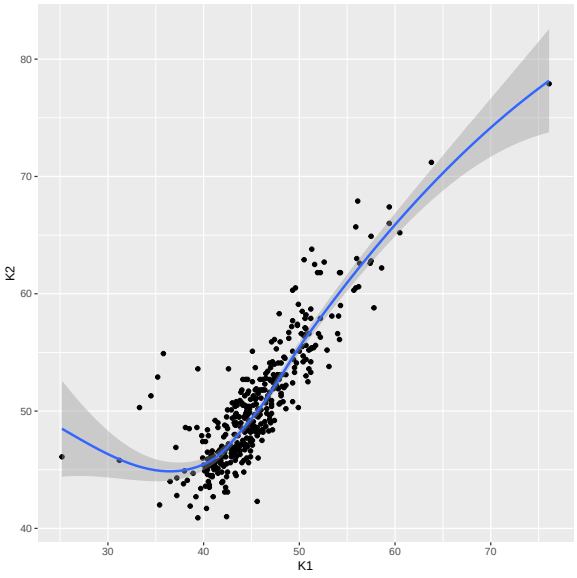


Figure 1: Default LOESS smoother on K_1 vs. K_2.

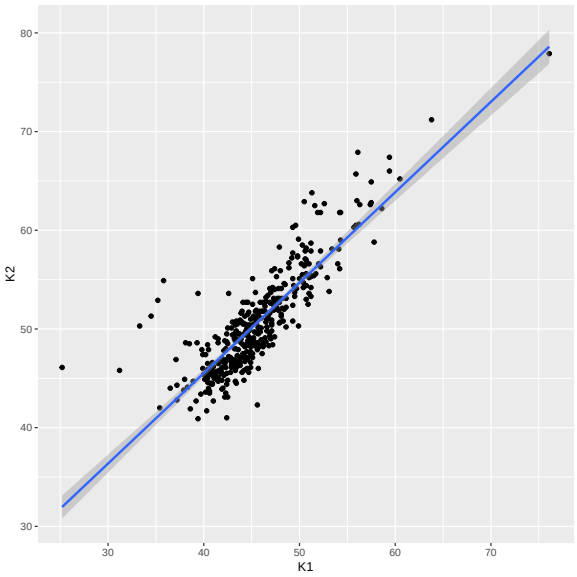


Figure 2: Linear model fit (`method = "lm"`).

The LOESS smoother (Figure 1) reveals a non-linear curvature flattening below $K_1 < 40$ diopters, transitioning into a strong linear regime across the bulk of the patient cohort ($40 \leq K_1 \leq 60$). At the extreme upper bound ($K_1 > 60$), the 95% confidence envelope widens significantly due to sample sparsity in severe advanced ectasias. Conversely, the OLS linear model (Figure 2) imposes a constant slope throughout the domain ($K_2 \approx \beta_0 + \beta_1 K_1$), confirming strong overall collinearity ($r > 0.85$) while obscuring localized deviations at low curvature boundaries.

3 Exercise 2: Keratometric Relationship Stratified by Ring Count

3.1 Objective and Implementation

We assess whether the morphological relationship between K_1 and K_2 exhibits structural divergence conditional on the surgical decision to implant either one or two rings (`na` $\in \{1, 2\}$).

```
p2 <- qplot(K1, K2,
            data = queratocono,
            colour = factor(na),
            geom = c("point", "smooth"),
            method = "lm",
            main = "Relation between K1 and K2")
```

3.2 Graphical Evaluation and Discussion

Mapping `colour = factor(na)` instructs `qplot()` to partition the regression estimator across levels 1 and 2 independently:

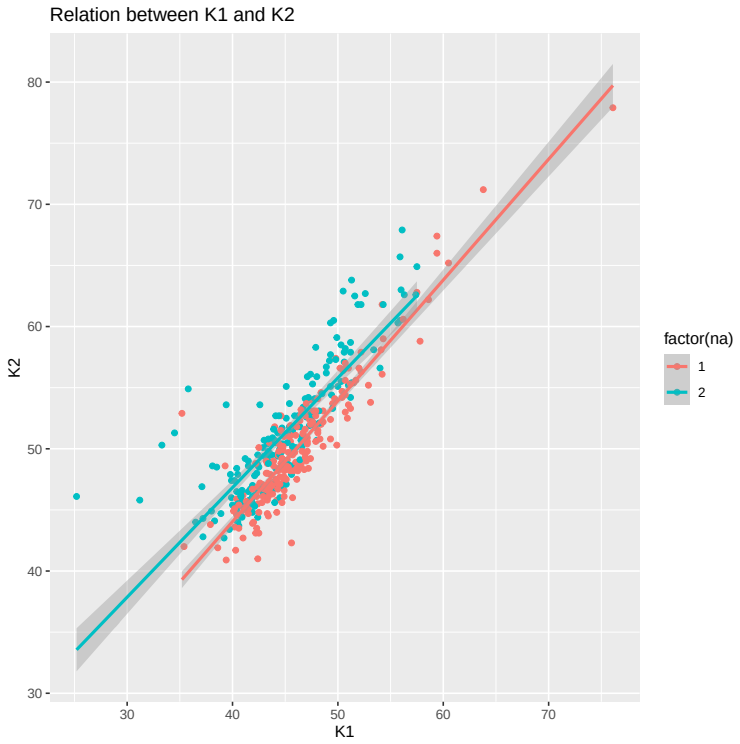


Figure 3: Stratified linear regressions for K_1 vs. K_2 by factor(`na`).

- **Single-ring implants (`na = 1`, red):** Patients receiving a single ring segment present a slightly steeper slope and higher dispersion in advanced curvatures, often corresponding to asymmetrical, highly localized inferior cones.
- **Double-ring implants (`na = 2`, blue/cyan):** Concentrated along a tighter trajectory, reflecting central, symmetric keratoconic cones where dual segments are indicated to achieve uniform circumferential flattening.

4 Exercise 3: Pre-operative vs. Post-operative Curvature

4.1 Objective and Implementation

The surgical efficacy of the procedure is evaluated by confronting pre-operative principal keratometry (K_1) with post-operative principal keratometry ($K_{1.\text{output}}$).

```
p3 <- qplot(K1, K1.salida, data = queratocono, geom = "point")
```

4.2 Graphical Evaluation and Discussion

The scatter plot exhibits an extremely tight linear correlation, demonstrating predictable surgical predictability across the sample. Most data points lie below the theoretical identity line $y = x$, visually confirming the desired flattening effect ($K_{1.\text{output}} < K_1$). Two notable clinical outliers are detected: an isolated patient displaying exceptional pre-operative steepness ($K_1 > 75$, remaining near 45 diopters post-operatively) and an atypical measurement near $K_1 \approx 64$ with $K_{1.\text{output}} \approx 63$, indicating a candidate refractory to standard flattening.

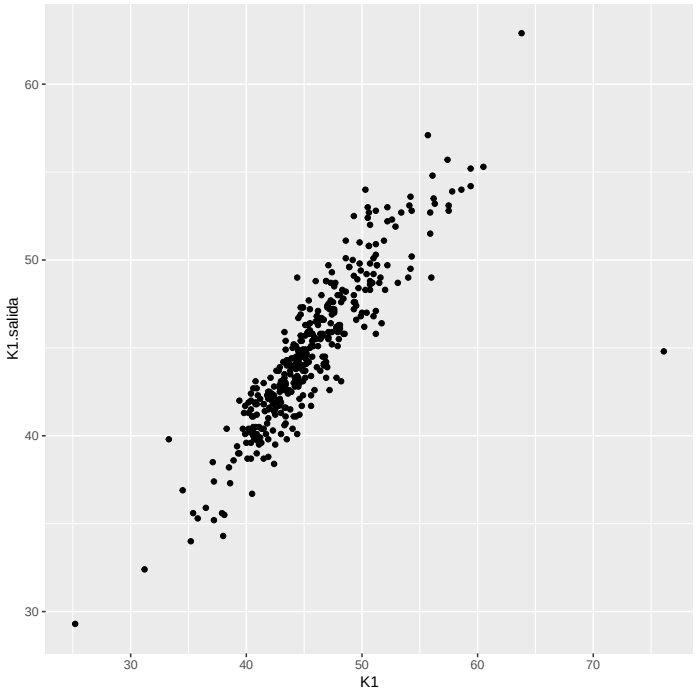


Figure 4: Pre-operative (K_1) vs. post-operative ($K_{1.\text{output}}$) corneal curvature.

5 Exercise 4: Distribution of Ring Thickness by Segment Count

5.1 Objective and Implementation

The frequency profile of ring thickness (`grosor`, in μm) is analyzed as a discrete factor, with the bar fill stacked according to segment count (`na`).

```
p4 <- qplot(factor(grosor), data = queratocono, geom = "bar", fill = factor(na))
```

5.2 Graphical Evaluation and Discussion

The empirical distribution indicates marked clinical preference for 200 μm ($n = 179$) and 250 μm ($n = 127$) rings. Intermediate thicknesses (175 μm, 225 μm, 275 μm) are deployed exclusively in dual-ring configurations (`na = 2`), where asymmetric combinations of different thicknesses are paired to correct asymmetrical astigmatic components.

6 Exercise 5: Multi-dimensional Faceting by Incision Diameter and Rings

6.1 Objective and Implementation

We examine high-dimensional interactions by conditioning the K_1–K_2 relationship across surgical incision diameter (`diam` $\in \{5, 6\}$ mm, row panels) and segment count (`na` $\in \{1, 2\}$, column panels), encoding ring thickness via color and mitigating point occlusion through an alpha transparency of 1/3.

```
p5 <- qplot(K1, K2,
            data = queratocono,
            facets = diam ~ na,
            colour = factor(grosor),
            alpha = I(1/3))
```

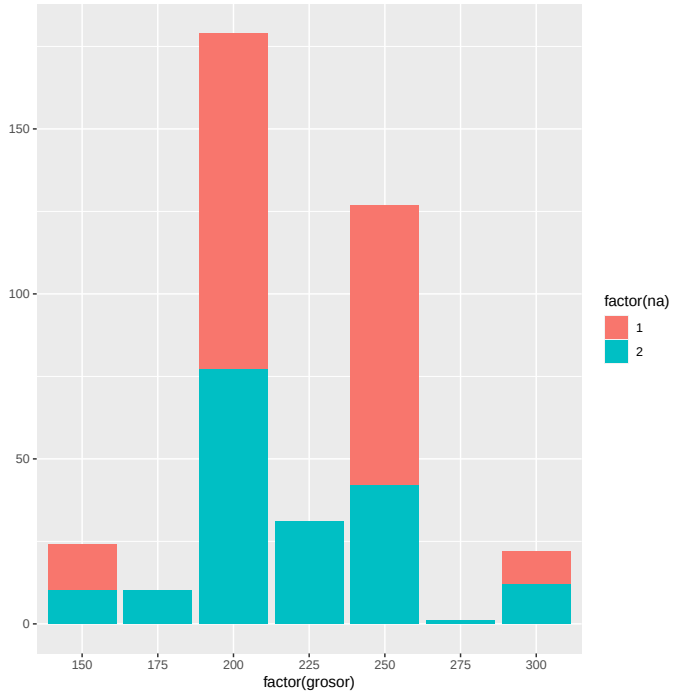


Figure 5: Stacked bar chart of factor(`grosor`) partitioned by factor(`na`).

6.2 Graphical Evaluation and Discussion

The faceting grid elucidates surgical protocol segmentation:

- Smaller optical diameter (5 mm, top row) is predominantly chosen for eyes with moderate keratometry, providing stronger focal refractive flattening closer to the visual axis.
- Larger diameter (6 mm, bottom row) exhibits broader corneal coverage, accommodating eyes with steeper baseline deformities.
- The transparency factor $\alpha = 1/3$ effectively reveals point density clusters around $K_1 \in [42, 48]$ and prevents misinterpretation of high-density overlap regions.

7 Exercise 6: Distributional Summaries via Boxplots

7.1 Objective and Implementation

To assess median curvature progression and interquartile dispersion across segment thicknesses, independent box-and-whisker plots are constructed for both corneal axes.

```
p6_k1 <- qplot(factor(grosor), K1, data = queratocono, geom = "boxplot")
p6_k2 <- qplot(factor(grosor), K2, data = queratocono, geom = "boxplot")
```

7.2 Graphical Evaluation and Discussion

Across both axes, median keratometric steepness rises monotonically with ring thickness: eyes requiring thicker rings (225 μm to 300 μm) exhibit higher baseline disease stages. The 200 μm category shows the highest number of positive upper outliers ($K_1, K_2 > 65$), reflecting its status as the most versatile standard choice across diverse disease profiles.

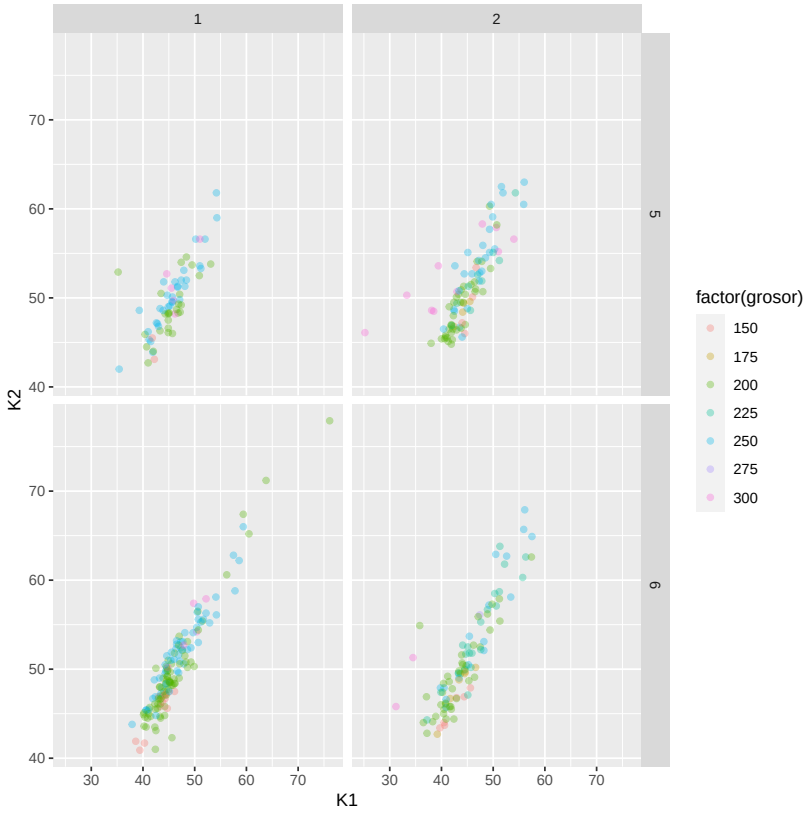


Figure 6: Faceted scatter plot (2×2 grid: `diam` $\sim$ `na`) with alpha blending.

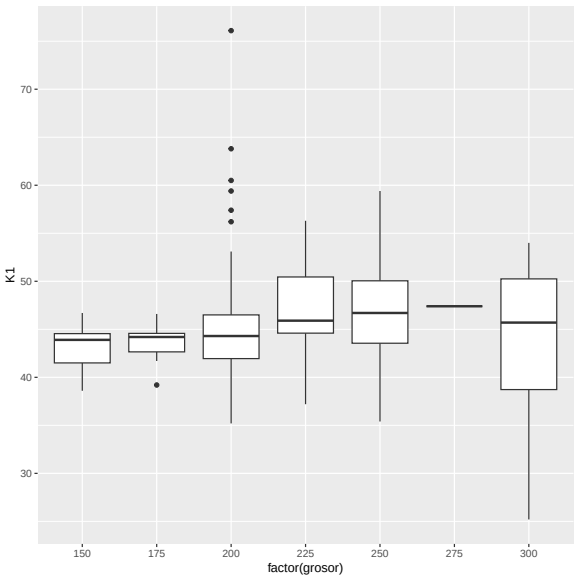


Figure 7: Boxplot of K_1 across `factor(grosor)`.

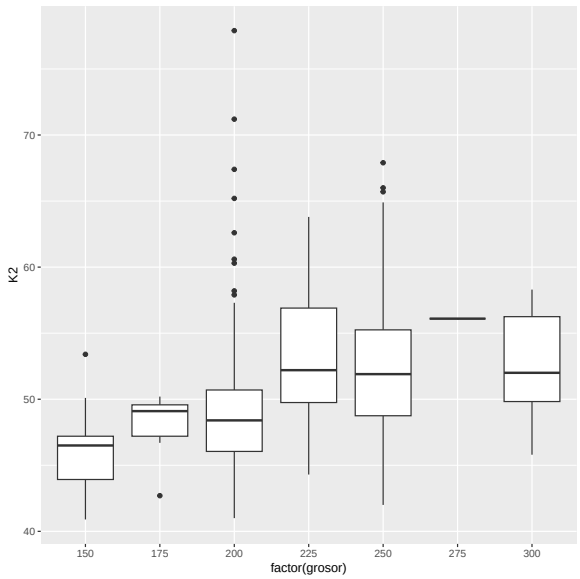


Figure 8: Boxplot of K_2 across `factor(grosor)`.

Appendix: Complete Sequential Execution Log

The verbatim console output recorded during the sequential execution of `pract3.R` is reproduced below:

```
'data.frame': 394 obs. of  16 variables:
 $K1       : num  38.1 39.4 43.3 45.7 44.6 44.2 44.2 53.1 45.1 40.7 ...$ K2       : num
     48.6 53.6 50.4 50.1 50.4 44.8 45.8 53.8 52.5 44.5 ...
 $ch       : num  6.9 5.7 8.8 11.1 11.1 8.9 11.1 7.5 7.7 9.4 ...$ na       : int  2 2 2
     1 2 1 1 1 2 1 ...
 $Incision : int  60 140 120 30 60 180 130 60 60 30 ...$ Prof     : int  390 400 381 370
     440 327 380 400 380 477 ...
 $diam     : int  5 5 5 5 5 6 6 5 6 5 ...$ grosor   : int  300 300 200 250 200 200 200
     200 225 200 ...
 $Longitud1: int  90 105 120 160 120 150 210 160 120 160 ...$ Longitud2: int  180 210
     240 160 240 150 210 160 240 160 ...
 $grosor1  : int  300 300 200 250 200 200 200 200 200 200 ...$ grosor2  : int  300 300
     200 0 200 0 0 0 250 0 ...
 $long1    : int  90 90 120 160 120 150 210 160 90 160 ...$ long2    : int  90 120 120 0
     120 0 0 0 150 0 ...
 $K1.salida: num  35.5 42 42.7 45.8 44.4 42.8 43.1 48.7 45.7 39.8 ...$ Astig    : num
     10.1 8.9 3 1.9 2.9 2.3 3.9 2.8 3.2 3.8 ...
    K1   K2   ch na Incision Prof diam grosor Longitud1 Longitud2 grosor1
1 38.1 48.6  6.9  2       60  390    5    300        90        180     300
2 39.4 53.6  5.7  2      140  400    5    300       105        210     300
3 43.3 50.4  8.8  2      120  381    5    200       120        240     200
4 45.7 50.1 11.1  1       30  370    5    250       160        160     250
5 44.6 50.4 11.1  2       60  440    5    200       120        240     200
6 44.2 44.8  8.9  1      180  327    6    200       150        150     200
  grosor2 long1 long2 K1.salida Astig
1     300    90    90      35.5  10.1
2     300    90   120      42.0   8.9
3     200   120   120      42.7   3.0
4       0   160     0      45.8   1.9
5     200   120   120      44.4   2.9
6       0   150     0      42.8   2.3
Warning message:
‘qplot()‘ was deprecated in ggplot2 3.4.0.
‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
pdf
  2
Warning message:
In geom_point(method = "lm") : Ignoring unknown parameters: ‘method‘
‘geom_smooth()‘ using formula = ’y ~ x’
‘geom_smooth()‘ using formula = ’y ~ x’
pdf
  2
Warning message:
In geom_point(method = "lm") : Ignoring unknown parameters: ‘method‘
‘geom_smooth()‘ using formula = ’y ~ x’
‘geom_smooth()‘ using formula = ’y ~ x’
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```